RFWNet: A Multiscale Remote Sensing Forest Wildfire Detection Network With Digital Twinning, Adaptive Spatial Aggregation, and Dynamic Sparse Features

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***Abstract*— Real-time detection of forest fires through remote sensing is a challenging task, especially in the context of limited data availability. In response to this challenge, this article leverages the digital twin (DT) concept to create a comprehensive and high-fidelity synthetic forest wildfire dataset. Alongside this, we have made available a high-resolution forest fire remote sensing dataset from real scenarios, meticulously collected and annotated by our research team. Aiming for precision in detecting forest fires via remote sensing, we present the remote sensing forest wildfire detection network (RFWNet) and its lightweight version, RFWNet-nano. More specifically, our network's backbone, grounded on deformable convolution network v3 (DCNv3), develops a multigroup mechanism, amplifying its ability to perceive the correlations overextended distances. Utilizing our dual-path dynamic sparse attention (DDSA), we meld coarse-grained regional selection with granular token-to-token attention, adeptly capturing the evolving contours of fires and smoke. To address diverse scenarios, our Vanilla Head design, backed by a profound training approach and simultaneous stacked activations, accurately identifies flames and smoke across multiple scales. Furthermore, we advocate for a 24/7 real-time monitoring system, synergizing drones, edge computing devices, and NVIDIA GPUs. Our experimental outcomes indicate that relative to numerous prevailing object detection algorithms, RFWNet and RFWNet-nano both manifest considerable superiority in terms of quantitative precision and visual results, substantiating the robustness and preeminence of our methodologies.**

***Index Terms*— Digital twinning in remote forest wildfire, dual-path dynamic sparse attention (DDSA), real-time forest wildfire monitoring system, remote sensing forest wildfire detection network (RFWNet), Vanilla Head in object detection.**

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I. Introduction

FORESTS have enormous potential in mitigating climate change and environmental changes [1]. However, escalating extreme weather events and human interventions have led to a marked increase in both the frequency and intensity of forest fires. According to data from the Ministry of Emergency Management of China, a total of 709 forest fires occurred in 2022, damaging approximately 4689.5 hectares of forest area [2]. Emissions of smoke and gases from forest fires not only contribute to air pollution but also have detrimental effects on human health, leading to conditions, such as respiratory ailments [3], [4], [5], cardiovascular complications [6], [7], [8], and among others. In addition, forest fires can wreak havoc on transportation, communication, and power infrastructures, endangering properties, agriculture, wildlife, and flora [9]. Therefore, rapid and accurate forest fire warnings are of paramount importance for the sustainable management and forest protection [10].

Early warning systems for forest fires offer prompt fire-related insights, aiding forest management entities in deploying effective countermeasures via capabilities such as continuous monitoring and instant alerts. Common early warning systems of forest fire include ground patrols, satellite remote sensing [11], [12], [13], and drone surveillance [14], [15], [16]. Ground patrols are limited by factors such as terrain and climate and are difficult to conduct in certain complex areas. Moreover, ground patrols require a significant amount of manpower and time and are difficult to achieve comprehensive coverage. While satellite remote sensing boasts expansive coverage, its real-time capabilities falter, and cloud interference diminishes its monitoring efficacy in some regions [17]. In contrast, drones adeptly navigate forested terrains, largely unaffected by challenges like topography and weather conditions. Drones offer instantaneous on-the-spot imagery and footage, accessing areas challenging for satellite sensing and ground patrols to cover [18]. Furthermore, drones can be

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equipped with various devices, such as infrared thermal imagers and hyperspectral sensors, providing richer fire information. By incorporating deep learning-based object detection algorithms, real-time and accurate monitoring of forest fires can be achieved in all weather conditions [19].

Dataset quality is pivotal to the precision of forest fire warnings. However, there remains a dearth of expansive, high-caliber remote sensing data on forest fires. Digital twins (DTs) harness real-world data across various disciplines, quantities, scales, and probabilities, digitally mapping them via state-of-the-art (SOTA) technology [20]. The quintessential DT model, encompassing the physical realm, virtual modeling, twin data, services, and components [21], finds applications in domains like intelligent manufacturing and smart cities. Guo et al. [22] applied urban DTs to road pothole detection, obtaining 1000 high-resolution images of road potholes and performing pixel-level annotation, effectively addressing the problem of insufficient high-quality road pothole datasets. Li et al. [23] created virtual datasets by using DT technology to overcome the lack of real samples, achieving the fusion of virtual and real. Therefore, we embrace DT technology to emulate real-world forest environments, producing expansive and superior remote sensing datasets on forest fires. Such datasets encompass a gamut of forest scenarios, varying fire magnitudes, and diverse meteorological conditions. Employing advanced rendering methods, such as unreal engine, allows to mirror intricate forest specifics and fire dynamics, broadening dataset diversity and addressing the paucity of remote sensing data on forest fires.

Remote sensing images of forest fires encompass diverse environments, such as forests, grasslands, mountains, hills, rivers, and lakes, all while facing complex weather variations. In the context of multiscale detection of flames and smoke, despite the availability of advanced methods, there still exist significant technical challenges and shortcomings.

1) *Long-Distance Correlation Issues:* When dealing with remote sensing images of fire, it is essential to address the long-distance correlations between fire and smoke across different locations and scales. Although the deep feature interaction network (DFINet) proposed by Gao et al. [24] effectively highlights the autocorrelation and cross correlation of features in wetland classification, its applicability to complex forest fire scenarios is limited. There is a need for a deeper exploration of feature interaction techniques specifically tailored to the dynamic nature and environmental changes of fires. Shi et al. [25] have enhanced YOLOv5 by incorporating dilated convolution and polarized self-attention, improving the model's capacity to handle long-distance dependencies, and thus boosting detection accuracy. Nonetheless, challenges persist in capturing long-distance correlations, especially in the context of rapidly spreading forest fires. Lin et al. [26] presented a semi-supervised forest fire detection method that integrates YOLOv5 with a Transformer. Although this approach enhances global information capture and localization accuracy, it raises questions regarding efficiency and accuracy in the real-time processing of large-scale fire scenes. Sun et al. [27] employed Mahalanobis distance for smoke detection in satellite imagery, which was suitable for known stable smoke characteristics. However, further enhancements are needed for precise and rapid detection of dynamically changing fire smoke, especially over long distances. Yang et al. [28] proposed a strategy for multitemporal video smoke detection that effectively addressed spatial and temporal dependency issues. However, when applied to real-time remote sensing data, particularly in managing the long-distance correlations in forest fire detection, the scalability and efficiency of the algorithm required further verification. Li et al. [29] conducted a study that enhanced smoke detection accuracy through random forests and subpixel mapping. While this approach improved smoke recognition, it encountered challenges in quickly and accurately identifying dynamically changing fires and smoke over long distances, especially in complex meteorological and terrain conditions.
2) *Issues With Neural Network's Feature Focus:* Hong et al. [30] proposed the shared and specific feature learning (S2FL) model, which effectively separated shared and specific modal components in multimodal remote sensing data, improving the accuracy of land cover classification. However, despite its advantages in integrating multisource data, the S2FL model still faced challenges in distinguishing subtle differences between features during practical implementation. Li et al. [29] introduced the ECA-DSNN model, which combined U-Net and Transformer architectures, aimed to enhance the capture of global and local features for building recognition. However, the model became overly sensitive to background features in remote sensing images with complex backgrounds, potentially compromising the accuracy of target recognition. Eftekhari et al. [31] proposed a method that employed parallel space-channel attention mechanisms to enhance feature discrimination. However, this heavy reliance on attention modules could lead to false detections in static or homogeneous backgrounds, as the model might overinterpret minor variations in the scene. Ren et al. [32] employed the feature constraint and local guided global feature model, which enhanced the object detection performance by constraining positive sample features and guiding global features locally. This model performed well in forest fire images obstructed by smoke or trees. However, over-optimization for specific features limited the model's adaptability and generalization capabilities in broader or unseen environments. Wang et al. [33] presented coupled CNN and Transformer network (CCTNet), which integrated CNNs and Transformers to enhance semantic segmentation for ground crop classification. However, when dealing with ultrahigh-resolution images, the method necessitated splitting images into small pieces.

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This led to incomplete edge structures of different categories and potential classification errors due to a lack of global information.

3) *Multiscale Flame and Smoke Detection:* While Zhang et al. [34] improved detection rates for objects of different sizes with multiscale convergence coordinated pyramid network with mixed attention and fast-robust NMS (MMFNet) by designing a multiscale fusion coordinated pyramid network with a hybrid attention mechanism, the system faced challenges in handling low concentration or very small-volume smoke. This suggests that existing multiscale processing strategies might have lacked comprehensiveness in fine-grained feature extraction, particularly in the early identification of smoke in fire scenarios. Jiang et al. [35] optimized multiscale feature fusion small object detection network (MFFSODNet) for small target detection in images captured by drones. They achieved this by designing multiscale feature extraction modules and a bidirectional dense feature pyramid network to integrate shallow fine-grained information with deep semantic information. However, this method could result in information loss when fused with deep features, especially in rapidly changing fire scenes, where this fusion strategy might not accurately reflect the fire's spread in a timely manner. Feng and Sun [36] proposed a feature fusion-based multiscale network for fire detection in complex scenes. Despite enhancing feature calibration and multiscale spatial feature pooling through dense connections and group channel attention, the model had limitations in handling scale variations caused by changes in fire size. This highlights the shortcomings of existing fire detection methods in adapting to dynamically changing fire scales and real-time processing requirements.

In response to these challenges, we developed a novel remote sensing forest wildfire detection network (RFWNet). This network employs drones, edge computing devices (Nvidia Jetson Xavier NX), and NVIDIA GPUs to facilitate round-the-clock forest fire monitoring. The main contributions of this article are as follows.

1) Addressing the scarcity of remote sensing data for forest fire detection, this article combines DT technology with actual UAV-collected data to generate a comprehensive synthetic dataset (M4SFWD) and a real-world dataset from Yuxi, China, enhancing data resources for remote sensing of forest fires.
2) We introduce an innovative backbone network based on DCNv3 with a dual-path dynamic sparse attention (DDSA) mechanism and a Vanilla Head design, optimizing the capture of dynamic shapes and structures of fire and smoke and improving detection accuracy in diverse forest fire scenarios.
3) A comprehensive real-time monitoring system for forest fires is proposed, integrating RFWNet with drones, edge computing devices, and NVIDIA GPUs, ensuring efficient all-weather surveillance and adaptability to various hardware capabilities.

The remainder of this article is organized as follows. Section II reviews related work, including remote sensing of forest fire detection, backbone, and attention mechanisms. Section III elaborately introduces the proposed methodology, encompassing the construction of datasets, the design of a real-time remote sensing forest wildfire detection system, and the network architecture of RFWNet. Specifically, we delve into the design of the DCN-based backbone network, the DDSA mechanism, and the Vanilla Head. Section IV presents the experimental results and discussion, including experimental setups, comparative experiments with the YOLO series of detection algorithms, comparisons with other object detection algorithms, and a visualization analysis of the experimental outcomes. Finally, Section V draws the conclusion, summarizing the contributions and suggesting directions for future work.

II. Related Works

A. Remote Sensing of Forest Fire Detection

Forest fires, characterized by their swift spread and immense destructiveness, pose grave threats to both the environment and human health. The prompt and efficient detection of forest fires has always been a research focus.

1) Traditional Forest Fire Detection Methods: The conventional fire detection techniques predominantly depend on manual feature extraction, particularly emphasizing the hues [37], [38], [39] and contours [40], [41] of fires and smoke. Çelik and Demirel [40] proposed a generic color model for flame pixel classification based on YCbCr color space, achieving a fire detection rate of up to 99%. However, this method struggles with varying lighting conditions and is overly dependent on the choice of color space. Merino et al. [42] demonstrated an unmanned aerial vehicle (UAV) system for automatic monitoring of forest fires, capturing real-time information of the fire front through infrared or visible cameras. Despite its capability to cover extensive areas, the computational complexity is high. Chen et al. [43] introduced an early fire alarm method based on the RGB model's color and disorder metrics; yet, it requires further optimization for dynamic flames and complex backgrounds. Töreyin et al.'s [44] study detected flame flicker through video data from standard cameras analyzed in the wavelet domain, effectively reducing false alarms. However, the algorithm's complexity makes it unsuitable for resource-constrained environments. Giglio et al. [45] introduced an enhanced fire detection algorithm for the Moderate Resolution Imaging Spectroradiometer (MODIS), improving sensitivity to tiny fires and significantly reducing the false alarm rate. However, this method, designed for a specific platform, has limited its applicability. In summary, while traditional forest fire detection technologies have made some progress in improving the accuracy and timeliness of fire detection, these methods suffer from poor robustness, being easily affected by environmental lighting, which leads to a high rate of false alarms. Furthermore, the intricacies of manual feature design necessitate intricate and labor-intensive feature engineering, challenging the balance between accuracy and timely execution.

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2) Deep Learning Based Forest Fire Detection Method: Recently, researchers have proposed fire detection methods based on neural networks to address these issues. The archetypal convolutional neural network (CNN), LeNet5, when employed for fire detection [46], [47], has outperformed conventional methods in terms of accuracy. However, this approach overlooks the model's computational intricacy, and an extensive model size proves challenging for devices with limited resources. Consequently, researchers have delved into object detection models like faster R-CNN [48] and YOLO [49], [50], [51], [52], aiming for superior performance with reduced model complexity in fire detection scenarios.

Object detection methods based on deep learning can automatically extract fire features from a large amount of data, providing significant advantages in terms of real-time performance, adaptability, and scalability. This effectively addresses the real-time requirements, accuracy, and robustness of forest fire detection. Wu and Zhang [53] utilized three prevalent object detection architectures (faster R-CNN, YOLO, and SSD [54]) for flame and smoke discernment, attaining instantaneous forest fire detection. Aiming to enhance real-time efficacy, Wang et al. [55] integrated MobileNet with YOLO v4, facilitating streamlined forest fire detection on the NVIDIA Jetson Xavier NX edge computing platform. Abdusalomov et al. [56] proposed an automated forest fire detection method based on Detectron2 and mask R-CNN, improving the detection accuracy of forest fires by customizing the fire dataset and using data augmentation techniques. However, these approaches predominantly employ standard object detection models, neglecting tailored network designs specifically for capturing flame and smoke characteristics. Consequently, notable potential remains for refining the accuracy of these models in detecting flame and smoke. To address this issue, Bahhar et al. [57] introduced a phased fire detection schema integrating CNN. This method capitalizes on adaptive image scaling and a specialized backbone network, ensuring localized cross-layer integration and garnering a holistic grasp of smoke attributes. However, this method can only detect smoke while cannot extract flame features. Yar et al. [58] proposed a scale-layered detection approach by combining the focus module of the stem with spatial pyramid pooling to detect large and small fire areas. Saydirasulovich et al. [59] employed both single-channel and multichannel fusion techniques, targeting flame detection at minor scales and smoke detection at standard scales. However, these techniques are tailored primarily for detecting fire attributes at ground level.

3) Forest Fire Detection Method Based on Remote Sensing Technology and Deep Learning: The background of remote sensing forest fires is complex, with early-stage flames and smoke being extremely small in scale and susceptible to atmospheric conditions, cloud cover, lighting, and ground reflection, posing a classic challenge in detecting tiny targets via remote sensing. Conversely, the scale of flames and smoke in the later stages is significantly larger. Therefore, enhancing the model's accuracy in detecting multiscale flame targets remains a hot topic and challenge in the current research. Guede-Fernández et al. [60] introduced a deep learning-based object recognition system capable of identifying smoke columns below or above the horizon, yet this method struggles to distinguish smoke and clouds in complex environments, especially at long distances. Abdusalomov et al. [56] improved forest fire detection accuracy to 99.3% through an enhanced Detectron2 model and deep learning approach, though the performance of this method in nighttime or low-visibility conditions has not been thoroughly explored. Zhang et al. [61] proposed an automated fire detection framework utilizing Sentinel-2 imagery, capable of processing approximately 12 000 km^2 of data in less than 6 min. However, the adaptability of this method to early microscale forest fire scenarios requires further validation. Zheng et al. [62] assessed a real-time smoke detection method based on deep CNNs, achieving a detection speed of 27 frames/s (FPS) with the YOLOv3 algorithm, yet its capability to detect fire targets of varying scales needs optimization. Dewangan et al. [63] introduced the FIgLib dataset and SmokeyNet model, specifically for real-time wildland fire smoke detection, but this approach still has room for improvement in detecting early fires and extremely microscale targets. Reis and Turk [64] detected forest fires using deep CNNs and a transfer learning approach, achieving 97.95% accuracy on the forest luminosity airborne-based machine learning evaluation (FLAMLEV) dataset with pretrained networks. However, the generalizability of this method in unknown environments remains an issue. Sudhakar et al. [65] proposed a UAV-based forest fire detection and monitoring system that reduced false alarms, but it faces challenges in adaptability to fire characteristics and real-time data processing capability in complex environments.

In summary, although remote sensing technology has made progress in forest fire detection, current methods still share common drawbacks in improving the detection accuracy of multiscale fire targets, such as adaptability to complex environments, nighttime and microscale target detection capabilities, and generalizability in new environments. Therefore, there is a need to continue exploring more efficient and adaptable models to enhance the accuracy and practicality of early fire detection.

B. Backbone

The backbone network serves as the core for the entire detection model. It is responsible for capturing the target's edge, texture, color, and other features from the input image, subsequently forwarding them to the detection head for precise localization and classification. Prominent large-scale backbone architectures encompass AlexNet, GoogLeNet, VGGNet, and ResNet, whereas their small-scale counterparts feature SqueezeNet, MobileNet, and ShuffleNet. Predominant backbones in object detection, exemplified by VGG16 in faster R-CNN, Darknet, and CSPResNet in the YOLO series, and EfficientNet in EfficientDet, have their roots in these standard architectures. These networks exhibit marked robustness when detecting conventional objects in real-world scenarios. However, when tailored for remote sensing forest fire detection, they grapple with challenges, including the detection of small-fires and smoke, adaptive spatial clustering, and

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limited long-range correlation. Wang et al. [66] proposed the Hourglass-shaped CSPResNet to expand the receptive field of the detection model, thereby improving the detection accuracy of small-scale targets. However, this method overlooks the correlation between small targets of different categories. Yar et al. [58] integrated the Stem module into the backbone network of the fire detection model to enhance the perception of the spatial information of small-scale flames and smoke by the focus layer. Ghali et al. [67] proposed a hybrid CNN-visual transformer backbone network, which utilized the architecture of U-Net and the method of patch embedding to encode and decode location information, achieving superior performance under different brightness and smoke concentration conditions. However, this approach demands numerous parameters and necessitates advanced computing hardware. Li et al. [68] crafted a bidirectional recursive feature pyramid network (BiFPN) for smoke feature fusion and enhancement. By combining the features extracted by BiFPN with the initial features of the backbone network, the model focuses on the crucial features of interference objects, such as smoke and clouds. However, these methodologies fall short in overcoming the inherent limitations of standard convolution, specifically in long-range correlation and adaptive spatial clustering.

C. Attention Mechanisms

The attention mechanism allows networks to adaptively allocate weights based on differing input positions, concentrating on key locations to capture vital information. Such a mechanism has been widely adopted in both CNNs and visual Transformers. In the early-stage remote sensing of forest fires, detecting small targets is paramount. Incorporating attention mechanisms notably enhances the model's efficacy in identifying small-scale fires and smoke. Majid et al. [69] designed a global average pooling method to initially extract fire features, followed by highlighting the interested regions through gradient-weighted class activation mapping (Grad-CAM) for better visual recognition and localization by the detection head. Striving for a harmonious blend of real-time operation and precision, Yar et al. [70] proposed a dual-fire attention network. The channel attention first highlights the most important features from the backbone model's features, and then, an improved spatial attention mechanism captures the spatial details of the interested target, enhancing the model's detection capability for flames. Zhang et al. [34] proposed a hybrid attention-based multiscale coordinated convergence pyramid network to enhance the model's detection capability for low-density and small-volume fire targets. However, this methodology lacks robustness, rendering it vulnerable to natural factors such as lighting and fog. Wang et al. [71] proposed a self-attention and self-collaboration fire detection network that emphasizes the effective features of smoke through a lightweight self-attention mechanism. However, a crucial attribute of the conventional attention mechanism is the global receptive field, which calculates the weighted sum and feature interaction for all spatial positions, resulting in a high computational burden [72]. Given the stringent real-time demands of remote sensing in forest fire detection, the pursuit of more streamlined attention mechanisms emerges as a pivotal research trajectory. Several researchers have suggested integrating sparse attention [73] with visual transformers [74] as attention strategies [75], [76], enabling distinct queries to concentrate on specific key–value subsets. Furthermore, some studies have tackled the computational intricacy of traditional attention, deploying strategies like sparse connection paradigms [73] and low-rank approximations [77]. Zhu et al. [72] proposed a solution called dual-routing attention, which achieves content-aware sparse patterns in a query-adaptive manner through a region-level routing step and a token-level attention step. Therefore, to augment the model's adaptivity in recognizing flame and smoke regions and to leverage sparsity for heightened computational efficiency, we devised a DDSA mechanism.

III. Methodology

In this section, we discuss both the virtual forest fire dataset derived from remote sensing via DT technology and the actual fire dataset. Furthermore, we detail the real-time remote sensing system for forest fire detection and elaborate on the architecture of RFWNet.

A. Datasets

Our research work utilizes two primary datasets: 1) synthetic dataset and 2) real-world dataset. Specifically, the real-world dataset comprises the fire dataset made publicly available by Wu et al. [78], as well as a wildfire dataset collected by our research team in Yuxi, Yunnan, China, in 2023. Fig. 1 illustrates the distribution and quantity of these datasets.

As depicted in Fig. 1, the synthetic dataset contains 3946 images, the public fire dataset encompasses 11 303 images, and the wildfire dataset, which we gathered from Yuxi City in Yunnan Province, consists of 876 images. Collectively, these datasets offer a total of 16 125 images for our fire detection study. Significantly, these images span diverse scenarios and conditions, offering a comprehensive view of various fire situations, which bolsters the model's ability to generalize. In terms of bounding box statistics, the synthetic dataset incorporates 17 613 bounding boxes, the open fire dataset has 25 038 bounding boxes, and the Yuxi wildfire dataset accounts for 4423 bounding boxes. This suggests that our datasets are richly annotated, supplying ample ground truth data for the model to discern fire characteristics. Of significance is that, across all datasets, 31 079 bounding boxes are labeled as "Fire" and 15 995 as "Smoke." These figures elucidate the rough distribution of flames versus smoke in the datasets, underscoring the challenges inherent in gathering and annotating smoke data.

For the training and validation of our model, we partitioned the dataset of 16 125 images into 12 393 for training and 3732 for validation. This division strategy guarantees sufficient data availability for model training. The training set comprises 35 628 bounding boxes, and the validation set has 11 446, indicating the dataset's richness for both the training and validation phases. Given the inherent challenges and intricacies of fire detection, these plentiful annotations fortify the model's training across a variety of scenarios and conditions.

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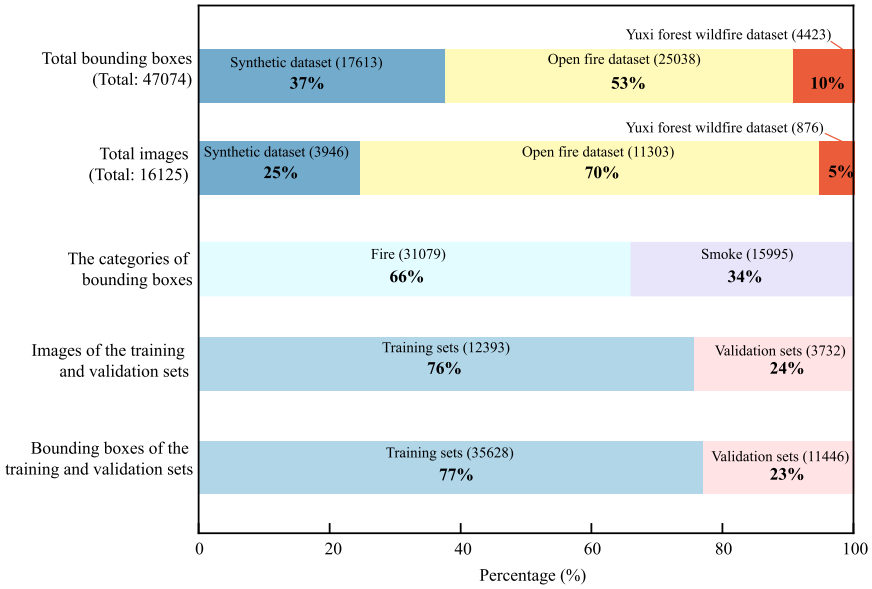


Fig. 1. Distribution of image quantities, bounding boxes, training, and validation sets employed in our study.

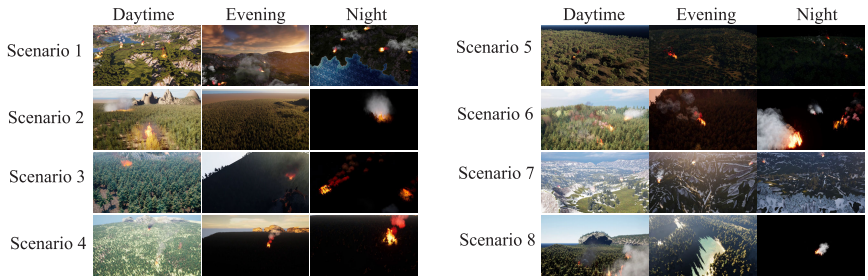


Fig. 2. Representative images from the synthetic dataset developed by leveraging DT technology. The dataset showcases eight unique terrain types, each under three lighting variations: daytime, dusk, and nighttime, and features a diverse count of flame and smoke instances.

1) M4SFWD: A Synthetic Forest Fire Dataset Employing Digital Twin Technology: We developed the M4SFWD, a synthetic fire dataset, harnessing the potential of DT Technology to simulate forest fires. As illustrated in Fig. 2, the dataset covers eight unique forest terrains—including lakes, mountains, and plains—accompanied by diverse weather patterns (from sunny to rainy/foggy and snowy days) and a range of light intensity levels (spanning daytime, dusk, and nighttime). Cumulatively, these datasets offer a drone's-eye view of remote sensing forest fire scenarios, consisting of 3974 images. With meticulous precision, our research team annotated the dataset at the pixel level, amounting to 17 763 bounding boxes, distinctly categorized as either fire or smoke. The efficacy of the synthetic dataset M4SFWD in enhancing the accuracy of remote sensing forest fire detection has already been established through our team's prior research. Detailed analysis and results, demonstrating the practical utility of M4SFWD in various detection scenarios, have been published in our previous work [79]. Consequently, this study does not include separate experiments to revalidate the effectiveness of the synthetic dataset.

Fig. 3 delineates the statistical attributes of the dataset. As portrayed, when viewed from a remote sensing lens, fires, and smoke from forest fires manifest unique spatial distributions and size profiles. Initially, the image count ratio between training and validation sets is 2.45:1, with the bounding box ratio being 2.5:1. On average, each image incorporates 4.5 bounding boxes, underscoring our detailed annotations. Moreover, the dataset prominently features bounding boxes of smaller sizes. Specifically, bounding boxes with dimensions less than 0.05 in both width and height constitute 29.8% of all annotations. In addition, bounding boxes measuring up to a width of 0.1 and a height of 0.15 cover a significant 90.8% of all annotations. This data underscores that predominant fire features, such as flames and smoke in remote sensing images, are localized and fine-scaled. Exploring the dimensions of flames and smoke further, the majority of flame-bounding boxes lie within sizes below 0.075, whereas smoke demonstrates a more expansive distribution, predominantly in the medium-size spectrum. This indicates that flames predominantly appear as bright, focused hotspots, while smoke, inherent to its diffusive quality, spans a broader spatial extent. Moreover, the heatmap representing bounding box centroids unveils a pivotal spatial trait: annotations intensify when the vertical axis falls below 0.7. The concentration of flame and smoke features in the lower half of remote sensing images account for this, reflecting the vertical propagation nature of flames and smoke.

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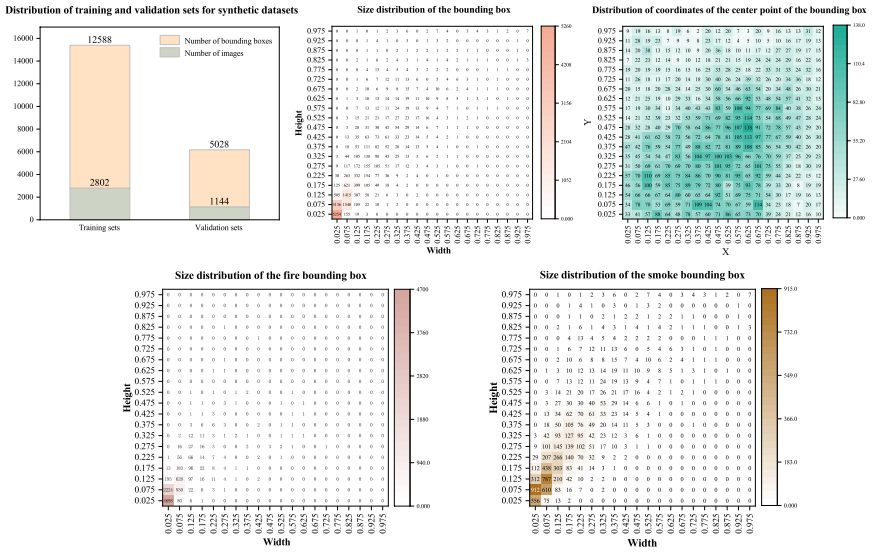


Fig. 3. Statistical attributes of our synthetic forest fire dataset.

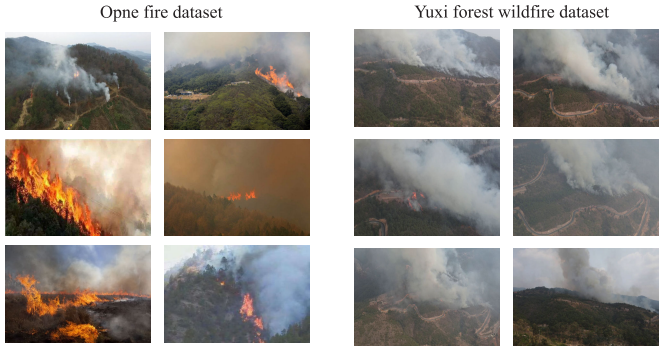


Fig. 4. (Left) Selection from the real fire dataset. (Right) Captured images of the Yuxi wildfire by our team.

2) Real Fire Dataset Analysis: Our adopted real-fire dataset comprises two parts.

1) The dataset presented by Wu et al. [78] comprises 11 303 fire images, encapsulating 25 038 annotated bounding boxes, as depicted in Fig. 4. It stands as a pre-eminent dataset for fire detection, encapsulating diverse fire scenarios from urban to wilderness landscapes. Such diversity augments the model's adaptability across varied scenes. Nonetheless, this dataset is not without its limitations. Given that a significant portion of the images originates from ground views, potential issues in adaptability may arise for aerial or high-altitude fire detection tasks. Moreover, terrestrial images are prone to disruptions from obstacles or fluctuating light conditions, potentially diminishing the clarity of flame and smoke attributes. Acknowledging these hindrances, we posit that relying solely on one data source is suboptimal for remote sensing forest fire detection. Consequently, we integrated a synthetic dataset and amalgamated it with on-site wildfire data from the Yuxi forest wildfire, curated by our research team, ensuring the model benefits from a richer, more varied data.
2) Our research team collected and annotated the Yuxi wildfire dataset. On April 11, 2023, Jiangchuan District of Yuxi City, Yunnan Province(Longitude: 102.70875 and Latitude: 24.42576) experienced a forest wildfire. Responding swiftly, our team reached the scene and utilized a DJI Mavic Air 2 drone, capturing 14 high-definition videos of the wildfire at 60 FPS and a resolution of 3840 × 2160. From these recordings, we derived 876 ultra-HD remote sensing images of the forest fire, as illustrated in Fig. 4. With careful annotation at the pixel level, we identified 4423 bounding boxes within these images. The dataset is available for access on GitHub (https://github.com/Philharmy-Wang/yuxifire).

Fig. 5 shows a comprehensive overview of the primary statistical features of the remote sensing forest fire dataset. Upon assessment of the data distribution, it is evident that there are 8966 images in the training set and 2337 in the validation set. These datasets comprise 19 856 and 5182 annotated boxes, equating to an average of 1.3 bounding boxes for each image in the training set and 2.2 in the validation set. Furthermore, the heat map representing bounding box dimensions reveals that the majority of fire characteristics in remote sensing images, such as flames and smoke, predominantly occupy smaller regions. The bounding box dimensions for fires are mainly clustered within a compact range, contrasting with smoke which shows a broader spread, mirroring the intense and localized nature of flames versus the diffuse spread of smoke. Given that a significant portion of fire images originates from personal devices like smartphones or cameras, there is an inherent inclination to position the fire centrally, resulting in a more concentrated heat map distribution around the image center.

Fig. 6 presents a detailed analysis of the statistical attributes of the forest wildfire dataset, which our team collected on April 11, 2023, from Jiangchuan District, Yuxi City, Yunnan Province. Derived from an actual forest fire in the locale, this dataset is assembled from 14 high-definition videos captured using a DJI Mavic Air 2 drone. From these recordings,

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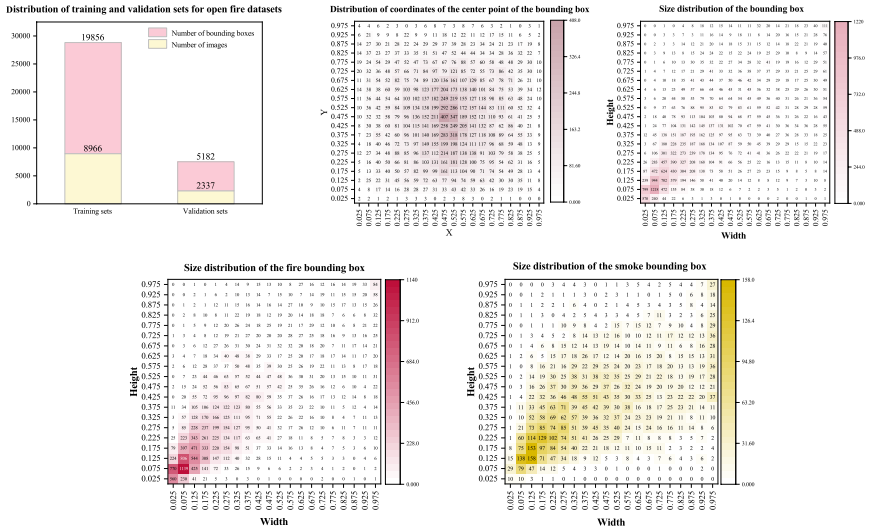


Fig. 5. Descriptive statistics of the publicly shared fire dataset.

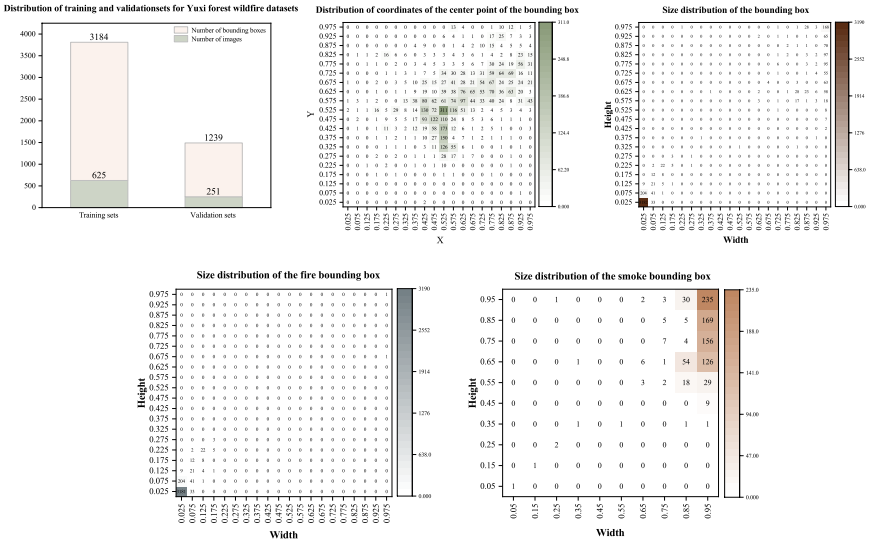


Fig. 6. Statistical characteristics of the Yuxi wildfire dataset.

876 ultra-HD remote sensing images were extracted, each having a resolution of 3840 × 2160. The data distribution illustrates that the training set includes 625 images with 3184 bounding boxes, averaging roughly 5.1 bounding boxes per image. Conversely, the validation set consists of 251 images with 1239 bounding boxes, averaging around 4.9 bounding boxes per image. Examining the heatmap for bounding box dimensions, it can be observed that the majority of the fire features, particularly flames, occupy smaller dimensions. In contrast, smoke generally expands more, leading to larger bounding box sizes. Recognizing the significance of this dataset, we have made it available on GitHub to benefit the wider research community.

B. Real-Time Remote Sensing Forest Fire Detection System

We designed two distinct versions of RFWNet, namely, RFWNet and RFWNet-nano. Tailored for high-performance computing devices, RFWNet features an expansive and intricate backbone network, employing four detection heads to accommodate fire and smoke targets across varied scales. Conversely, RFWNet-nano boasts a more streamlined network architecture, optimized for smaller image processing. This design facilitates real-time forest fire detection capabilities on edge computing platforms.

Fig. 7 illustrates a five-step conceptual model for forest fire prevention and alert. In Step 1, UAVs cruise predefined areas under all weather conditions. These UAVs, equipped with GPS positioning, also incorporate various imaging tools, including variable-focus visible light and infrared cameras, tailored to specific needs. The cruising zone has a designated return point for UAV communications, real-time image relay, battery substitution, and other functionalities. During Step 2, the UAVs' captured real-time video and flight data are resized to two dimensions: 448 × 448 and 832 × 832, streamlining processing across diverse models

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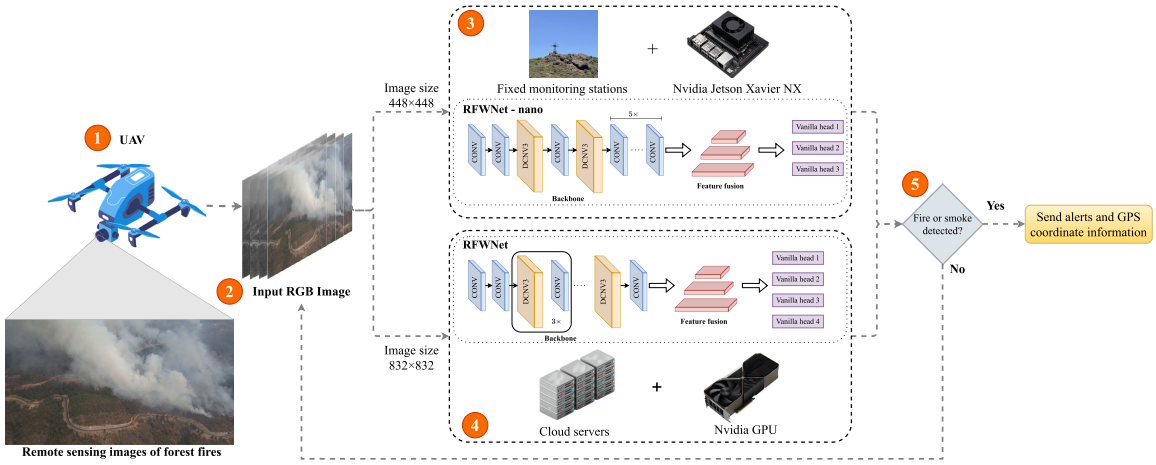


Fig. 7. Conceptual model proposed for forest fire detection via remote sensing.



Fig. 8. Lightweight real-time forest fire detection system based on RFWNet-nano.

and devices. Step 3 encompasses the establishment of permanent monitoring stations, with Fig. 7 showcasing one at the Heijinghe Nature Reserve in Kunming, Yunnan, China (longitude: 103.0661811° and latitude: 25.6159093°). This station, solar-powered, features cameras, alarms, and wireless communication tools but eschews high-end computational units. Hence, we utilize efficient edge computing devices, like Nvidia Jetson Xavier NX, and the compact RFWNet-nano to ensure continuous remote forest fire surveillance irrespective of weather. In Step 4, RFWNet processes the real-time 832 × 832 image/video feeds. Given the extensive input and model parameters, deployment on a cloud server becomes essential for computations. The choice in this phase can adjust to the real-time video's wireless transmission speed. Should a fire or smoke signature be identified in Step 5, the embedded systems dispatch a fire alert accompanied by the incident's GPS coordinates. Absent any fire detection, the sequence reverts to Step 2, maintaining area surveillance.

To further enhance the practicality and application scope of real-time remote sensing forest fire detection, we have constructed a lightweight real-time forest fire detection system based on RFWNet. The system primarily consists of a remote controller, video capture cards, an UAV, outdoor power sources, and other key hardware components, demonstrating the system's feasibility and convenience of deployment, as shown in Fig. 8.

In terms of hardware configuration, we have selected components suitable for field environments and capable of withstanding harsh weather conditions. As illustrated in Fig. 8(a), the drone is a lightweight yet powerful DJI MAVIC series, equipped with high-resolution cameras capable of transmitting high-quality image data in real time. Furthermore, we have also equipped the system with a portable outdoor power source to facilitate all-weather monitoring, ensuring stable operation even in remote areas without electrical grid coverage. The remote controller, a crucial device for operating the UAV, not only requires a user-friendly human–machine interface but also stable signal transmission capabilities. The video capture card transmits the video stream received from the UAV's camera to the Nvidia Jetson Xavier NX in real time. Fig. 8(b) displays the system in operation, where the UAV is mounted with video capture equipment, transmitting the collected images to the ground monitoring screen via wireless signals. Ground personnel can instantly view the images and swiftly respond based on the analysis results provided by our RFWNet-nano. This lightweight design allows RFWNet-nano

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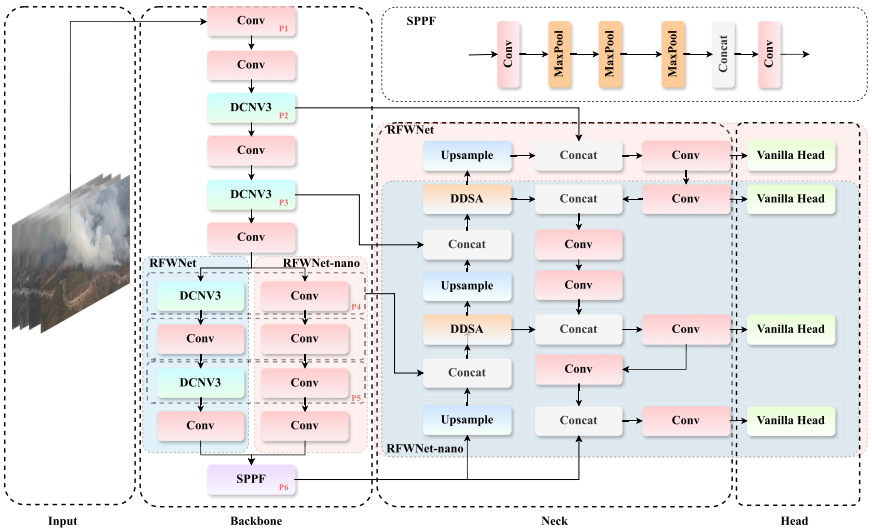


Fig. 9. Proposed RFWNet and RFWNet-nano, which contain four parts: input, backbone, neck, and head.

to operate in real time on edge computing devices such as the Nvidia Jetson Xavier NX, ensuring timely and accurate detection. Moreover, the system's portability and deployment ease mean that monitoring points can be quickly established in high-risk fire zones, significantly improving the spatial coverage and temporal responsiveness of forest fire prevention.

C. RFWNet

In this article, we propose RFWNet, a network for real-time detection of forest fires in the vision of remote sensing, as shown in Fig. 9. The RFWNet consists of four parts: input, backbone, neck, and head. For the backbone, we embrace the architectural foundations of the ResNet series and further augment it with a DCN-based backbone, integrating DCNv3. This fusion aims to discern long-range dependencies between fires and smoke in the imagery, simultaneously addressing the inherent constraints of traditional convolutions for adaptive spatial aggregations. The backbone for RFWNet amalgamates 4 $\times$ DCNv3 with conventional convolution, fortifying the extraction of profound and intricate feature sets. Contrarily, the RFWNet-nano's backbone merges 2 $\times$ DCNv3 and convolution, optimized with fewer parameters to fit lightweight detection schemes.

The multiscale feature fusion technique is designed to address the challenges of detecting targets of varying sizes. Specific scales of feature maps, such as P2–P5, are chosen to capture target characteristics ranging from small to large. This selection has been proven effective in multiple SOTA object detection architectures. The rationale behind choosing these levels is to leverage their unique receptive fields and detailed capturing abilities, thus optimizing the model's performance in detecting targets of varying sizes. Specifically, lower level feature maps (such as P2 and P3) provide the model with rich detail information beneficial for detecting small targets, while higher level feature maps (such as P4–P6) offer a broader contextual understanding, aiding in the recognition of larger targets. Through this strategy, our model is capable of maintaining real-time performance while enhancing the detection accuracy within the complex and variable datasets of remote sensing imagery.

The neck integrates a bidirectional feature pyramid, effectuating a top–down route via upsampling. This method methodically refines from semantically richer features down to rudimentary ones. Concurrently, the opposite is achieved through downsampling, collating, and deducing advanced features from the image's foundational ones. To enhance discernibility within flame and smoke regions, a DDSA mechanism is infused within the neck. Moving to the head, we present a minimalist approach inspired by VanillaNet, culminating in the Vanilla Head design. Its attributes such as scale, depth, and quantity are highly adaptable, contingent on model intricacies, detection environments, and target dimensions. Within this article, RFWNet-nano deploys three varied scales of Vanilla Head, apt for computation-restricted environments, while RFWNet employs four scales, catering to intricate detection scenarios.

The design of the RFWNet's loss function integrates category classification loss, bounding box regression loss, and DDSA loss to ensure the model's accuracy and efficiency in remote sensing forest fire detection tasks.

Category Classification Loss: The category classification loss employs cross-entropy loss, aimed at assessing the difference between the predicted category distribution and the actual labels. Given the predicted category probability distribution P and the actual label distribution T, the category classification loss can be expressed as follows:

$$L_{\text{cls}} = -\sum_{c=1}^{C_{\text{num}}} T_c \log(P_c) \tag{1}$$

where C_{num} is the number of categories, T_c and P_c are the actual label and the model's predicted probability for class c, respectively. This loss function promotes accuracy in predicting fire and smoke categories.

Bounding Box Regression Loss: The bounding box regression loss consists of distribution focal (DFL) loss and complete intersection over union (CIOU) loss. DFL loss addresses the

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bounding box regression problem in a discretized manner, transforming it into a series of binary classification problems. Its mathematical expression is as follows:

$$L_{\text{dfl}} = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{M} -\log\left(\frac{\exp\left(s_{ij}\left[l_{ij}\right]\right)}{\sum_{k=1}^{M}\exp\left(s_{ij}[k]\right)}\right) \tag{2}$$

where N is the number of positive samples, M is the number of discretized bounding box positions, s_{ij} is the predicted score for the ith positive sample corresponding to the jth bounding box dimension, and l_{ij} is the index of the actual value for the jth position of the ith sample. $s_{ij}[k]$ represents the predicted score for the ith sample at the jth position for the kth discrete value.

CIOU loss considers the overlap, the distance between center points, and the aspect ratio of bounding boxes to enhance the accuracy of bounding box predictions. It is expressed as follows:

$$L_{\text{ciou}} = 1 - \text{IoU} + \frac{\rho^2\left(b, g^{gt}\right)}{c^2_{l(b,g^{gt})}} + \alpha v \tag{3}$$

where IoU is the intersection over the union between the predicted and actual bounding boxes, $\rho^2(b, g^{gt})$ is the squared Euclidean distance between the centers of the predicted box b and the actual box g^{gt}, $c^2_{l(b,g^{gt})}$ is the diagonal length of the smallest enclosing box covering the predicted and actual boxes, v assesses the consistency of the aspect ratio between the predicted and actual boxes, and α is the weight factor for center point consistency.

Attention Mechanism Loss: To further enhance the model's capability to capture the features of fire and smoke, a DDSA mechanism is designed. The attention mechanism loss aims to optimize the model's focus on key areas, and its loss function can be simplified as follows:

$$L_{\text{DDSA}} = \frac{1}{Z}\sum_{n=1}^{Z} \|A_n - A_n^{gt}\|_2^2 \tag{4}$$

where Z is the total number of attention weight matrices, A_n is the nth predicted attention weight matrix, A_n^{gt} is the corresponding actual attention weight matrix, and $|\cdot|_2^2$ represents the Euclidean norm, measuring the discrepancy between the predicted and actual attention distributions. This loss encourages the model to allocate reasonable attention weights to the dynamic shapes and structures of fire and smoke, enhancing the model's discriminative ability.

In summary, the expression for the proposed total loss function is as follows:

$$L_{\text{total}} = \lambda_{\text{cls}}L_{\text{cls}} + \lambda_{\text{bbox}}(L_{\text{dfl}} + L_{\text{ciou}}) + \lambda_{\text{DDSA}}L_{\text{DDSA}}. \tag{5}$$

Here, λ_{cls}, λ_{bbox}, and λ_{DDSA} are the weight coefficients for category classification loss, bounding box regression loss, and attention mechanism loss, respectively, adjusting the contribution of each loss term in the total loss function. This approach enables to accurately classify and locate fire-related targets and effectively captures the long-distance relationships as well as the dynamic changes of fire and smoke, thereby enhancing the performance of remote sensing forest fire detection.

1) DCN-Based Backbone: In remote sensing-based forest fire detection, fires and smoke display a discernible correlation. Their spatial distribution is random, with the potential to intersect, coalesce, or scatter. Fires, when combusted, generate smoke, potentially leading to subsequent fires. Determining long-distance relationships between such targets and capturing distinct shape features remain formidable challenges. Conventionally, methods to tackle this issue have fallen into two primary categories.

1) Amplifying the model's ability to learn about the target by maximizing information retrieval about object dimensions and locations, often through data augmentation techniques. Yet, the efficacy of this strategy hinges on the judicious design and choice of augmentation methods. Ill-conceived data augmentation could infuse noise, thereby diminishing model performance.
2) Introducing methods like deformable convolutions [80], [81] and dilated convolutions [82] to mitigate the limitations of standard convolutions, including spatial invariance, confined receptive fields, and rigid object feature extraction.

By incorporating a deformation sampling grid, deformable convolutions [80], [81], [83] enable the convolution kernels to adapt their forms, effectively mitigating challenges linked to traditional convolution operations, such as shape adaptability and spatial invariance. This advancement enhances the efficacy of CNNs in intricate imaging scenarios.

For traditional 2-D convolution, let us denote the input feature map as x, and let R be a standard 3×3 grid, indicative of the convolution kernel. At position p_0, the output feature y aligns with the following equation:

$$y(p_0) = \sum_{p_n \in R} w(p_n) \cdot x(p_n + p_0) \tag{6}$$

where p_n represents the enumerated positions in R.

Standard convolutions, while effective for numerous computer vision tasks, encounter limitations in remote sensing forest fire detection due to their inherent spatial invariance and confined receptive field. These constraints impede their ability to accurately capture the spatial distribution of fire and smoke. In response, the deformable convolution network v1 (DCNv1) [80] emerged. Its foundational concept involves assigning an offset to each sampling point of the convolution kernel, as depicted in the following equation:

$$y(p_0) = \sum_{p_n \in R} w(p_n) \cdot x(p_0 + p_n + \Delta p_n) \tag{7}$$

where Δp_n signifies the offset discerned through an offset prediction network. This mechanism enables DCNv1 to adapt according to the actual morphology of fire and smoke, elevating detection precision.

Though DCNv1 achieves notable efficacy in delineating the intricate morphology of fire and smoke, certain constraints persist, notably its inflexibility in managing nonuniform deformations. To advance this technology, DCNv2 emerged [81], as depicted in the following equation:

$$y(p_0) = \sum_{p_n \in R} w(p_n) \cdot x(p_0 + p_n + \Delta p_n) \cdot \Delta m \tag{8}$$

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where Δm signifies the modulation scalar. Relative to DCNv1, DCNv2 not only offers a more sophisticated offset prediction but also integrates learnable sampling weights, amplifying the model's adaptability. These augmented features endow DCNv2 with amplified versatility in representing diverse morphologies of fire and smoke, thereby heightening the precision in remote sensing forest fire detection.

Though DCNv1 and DCNv2 excel in delineating the morphology of fire and smoke, they encounter difficulties with long-distance correlations and intricate spatial aggregations inherent in remote sensing forest fire images. In response, DCNv3 emerged. DCNv3 integrates an aggregation group mechanism, as depicted in the following equation:

$$\mathbf{y}(p_0) = \sum_{g=1}^{G} \sum_{p_n \in R} \mathbf{w}_g \mathbf{m}_{gn} \mathbf{x}_g \left(p_0 + p_n + \Delta p_{gn}\right) \tag{9}$$

where G signifies the count of spatial aggregation groups, and Δp_{gk} along with $\mathbf{m}_{gk}$ represents the sampling offset and modulation scale for each group, respectively. In contrast to DCNv2, DCNv3's distinct edge lies in its heightened proficiency in grasping the long-distance correlations within remote sensing images, a merit of its multigroup mechanism. Moreover, the adaptive spatial aggregation approach of DCNv3 bolsters its adeptness at discerning the dynamic morphologies of fire and smoke.

In the analysis of remote sensing forest fire images, the morphology and structure of fire and smoke undergo influences from factors including wind direction, topography, and types of vegetation. Such factors lead to both intricate and prolonged interconnections between fire and smoke. To adeptly grasp this sophisticated relationship, we present a backbone network built upon DCNv3, named the DCN-based backbone. The incorporation of its multigroup mechanism and an adaptive spatial aggregation approach provides a resilient and adaptable instrument, pinpointing the entire characteristics of fire and smoke, consequently amplifying the precision of remote sensing forest fire detection.

2) Dual-Path Dynamic Sparse Attention (DDSA): Initially, we re-examine the traditional attention mechanism. Given a query matrix Q, a key matrix K, and a value matrix V, attention $\mathcal{A}$ is derived as shown in the following equation:

$$\mathcal{A}(Q, K, V) = \text{soft}\max\left(\frac{QK^{\mathrm{T}}}{\sqrt{C}}\right)V \tag{10}$$

where C represents a scalar factor aiding in the mitigation of the vanishing gradient issue.

Within the standard attention mechanism, all tokens engage in key–value pairings. This engagement not only demands substantial computational resources but potentially incorporates extraneous information. In response to this challenge, several research endeavors [73], [76] have put forward diverse sparse attention methodologies. Nonetheless, a majority of these techniques depend on handcrafted sparse attention models, lacking the precision to discern the nuanced interconnections inherent in the data. To transcend these constraints and to effectively discern fire and smoke dynamics, we assimilate the principles of biformer [72] and propose DDSA.

In images acquired through remote sensing of forest fires, both the positioning and the structure of fires and smoke undergo frequent alterations, highlighting the need for an attention mechanism with adaptive responsiveness to these shifts. To cater to this requirement, we present the DDSA mechanism, adept at dynamically identifying the most illustrative fire and smoke zones within images while allocating unique attention weights to distinct areas.

DDSA's foundational principle revolves around the adoption of a bitiered routing approach. Initially, it dynamically identifies the regions of paramount relevance to a specific query at a broad scale, leading to a substantial reduction in the key–value pairs under consideration. Following this initial selection, the model delves deeper within these areas, administering granular token-to-token attention to guarantee the accurate extraction of pivotal details. Through the amalgamation of sparsity techniques with optimized matrix operations, DDSA not only manifests adaptive attention distribution but also amplifies computational efficacy.

a) Coarse-grained region selection: Given an input feature map denoted by $X \in \mathbb{R}^{H \times W \times C}$, we partition it into $S \times S$ distinct regions without any overlap, with each region encompassing (HW/S^2) vectors. This operation translates to reshaping X into the form $X^r \in \mathbb{R}^{S^2 \times (HW/S^2) \times C}$. In our pursuit to dynamically ascertain the region bearing the highest relevance to the specified query, we commence by procuring fresh representations for both query and key via linear projection, as depicted in the following equation:

$$Q = X^r W^q, \quad K = X^r W^k, V = X^r W^v. \tag{11}$$

Herein, W^q, W^k, and $W^v \in \mathbb{R}^{C \times C}$ stand for the weight matrices associated with the query, key, and value, respectively, facilitating the linear transformation of the provided data.

Subsequently, we determine the regional mean values for Q and K, which are designated as Q_r and $K_r \in \mathbb{R}^{S^2 \times C}$, respectively.

Following this, we establish an attention score matrix $A_r \in \mathbb{R}^{S^2 \times S^2}$ to gauge the association between regions, as illustrated in the following equation:

$$A_r = \text{soft}\max\left(\frac{Q_r K_r^t ext T}{\sqrt{C}}\right). \tag{12}$$

In the succeeding step, we harness the row-wise top-k operator to derive the index matrix, denoted by I^r, as presented in the following equation:

$$I^r = \text{topkIndex}(A_r). \tag{13}$$

The ith row of I^r contains the k indices most pertinent to the ith region, which are determined during the coarse-grained region selection phase. The value of k is determined based on the design requirements of the model and experimental needs, aiming to balance the attention focus with computational resource expenditure. In this article, the choice of k is defined through preliminary experiments that assess the impact of different k values on model performance. The objective is to identify an optimal balance, where the model effectively concentrates on key areas while maintaining computational efficiency. These preparatory trials have indicated that setting

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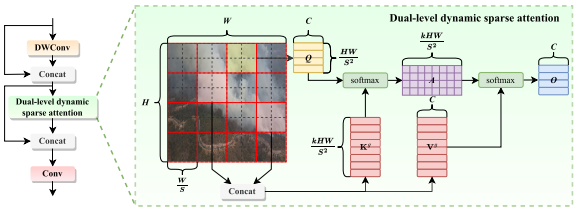


Fig. 10. Network architecture and computational graph of DDSA.

$k = 50$ allows the model to minimize computational resource consumption while ensuring performance; thus, $k = 50$ has been selected as the optimal value for this study.

b) Fine-grained token-to-token attention: Following the coarse-grained region selection, we further investigate these regions by harnessing the fine-grained token-to-token attention mechanism. The core objective of this mechanism lies in ensuring that each query token within our selected regions engages in more intricate interactions with other tokens. This strategy becomes crucial in discerning the intricate nuances of fire and smoke in remote sensing forest fire contexts.

In particular, for each query token situated within region i, our aim is to predominantly concentrate on tokens indexed by $I^r_{(i,1)}, I^r_{(i,2)}, \ldots, I^r_{(i,k)}$. Such indexed tokens emerge as the most pertinent to the query token based on our earlier coarse-grained selection. Given that these routed regions could be dispersed throughout the entire feature map, and considering modern GPUs favor processing continuous memory regions, we propose a novel gathering operation, as depicted in the following equation:

$$K^g = \text{gather}\left(K, I^r\right), \quad V^g = \text{gather}\left(V, I^r\right) \tag{14}$$

where $\mathbf{K}^g$ and $\mathbf{V}^g$ denote the tokens derived from the foundational key and value tensors, guided by the routing index matrix I^r. With these key–value pairs in place, we proceed to enact fine-grained attention, as delineated in the following equation:

$$O = \mathcal{A}\left(Q, K^g, V^g\right) + \text{LCE}(V). \tag{15}$$

We employ the local context enhancement (LCE) technique in this scenario. LCE() embodies a function steered by deep convolution. We attribute a kernel of size 5 to it, fortifying the accurate discernment of local features while addressing fire and smoke.

Fig. 10 demonstrates that by amassing key–value pairs from the k most pertinent windows, we sidestep computations in extraneous areas, concurrently assuring GPU-optimized processing through the embrace of dense matrix multiplication. This methodology proves especially advantageous for bolstering computational efficiency in the realm of large-scale remote sensing forest fire imagery.

Compared with traditional attention mechanisms, the proposed DDSA mechanism in this article demonstrates significant computational advantages and adaptability. First, standard attention models often employ fully connected operations, which could lead to excessive computational burden when processing large-scale remote sensing images. In contrast, DDSA optimizes the allocation of computational resources through its innovative dual-path structure. This mechanism not only swiftly identifies key features within a broader region but also concentrates processing power on these specific areas, thus significantly reducing unnecessary computational overhead. Second, unlike traditional models with limited applicability in complex environments, DDSA's design of dynamic sparse attention enables it to adaptively adjust its focus based on actual image content. This feature is crucial for rapidly changing scenes like fires, providing more accurate monitoring and analysis. Finally, by combining coarse-grained region selection with fine-grained token-to-token attention, DDSA effectively captures key features in multiscale environments, thereby enhancing the accuracy and efficiency of fire detection. This depth of analysis from macro to micro levels is difficult to achieve with traditional single-scale attention models, offering a more efficient and adaptable technical solution for remote sensing image analysis.

3) Vanilla Head: Within object detection architectures, networks of greater depth—owing to their elaborate layer configurations—tend to manifest enhanced nonlinearities, subsequently leading to superior outcomes across a range of tasks. Nonetheless, this depth invariably extends both training and inference durations. Striving for a balance between model simplicity and efficient performance, while curtailing network intricacy and the number of parameters, we embrace the streamlined philosophy behind VanillaNet and design the Vanilla Head. Central to the Vanilla Head is a depth training methodology harmonized with a tactic for simultaneous stacked activation functions.

a) Depth training strategy: At its core, this strategy entails deploying complex activation functions during the foundational stages of training two convolutional layers. As the training advances, this multifaceted activation function progressively transitions to an identity mapping.

For a specified activation function $\text{Act}(x)$, we pair it with an identity mapping. This association can be depicted as follows:

$$\text{Act}\prime(x) = \lambda x + (1 - \lambda)\text{Act}(x) \tag{16}$$

where λ serves as a hyperparameter, fine-tuned to balance the nonlinearity inherent in the activation function $\text{Act}\prime(x)$. To dynamically adjust this balance, we utilize the current training epoch e and the total training epochs E to determine ε, as shown in the following equation:

$$\lambda = \frac{e}{E}. \tag{17}$$

This configuration signifies that at the outset of training (with $e = 0$), $\text{Act}\prime(x) = \text{Act}(x)$, bestow the network with a distinct nonlinearity. However, as training culminates, $\text{Act}\prime(x) = x$, suggesting a potential fusion of the two convolutional layers, attributed to the missing activation function interlinking them.

Leveraging this methodology, the network benefits from a heightened nonlinearity in its nascent training phases, facilitating the discernment of intricate fire and smoke features. As the training progresses, the network incrementally streamlines, promoting swifter computational throughput during inference.

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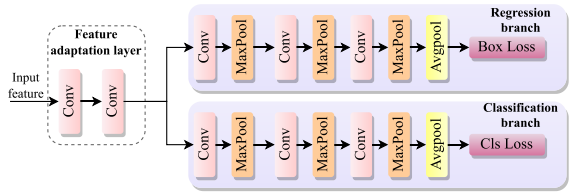


Fig. 11. Network architecture of Vanilla Head.

b) Stacked activation function: The principle behind concurrent stacked activation functions is to harness a diverse range of nonlinear features. While conventional deep learning models often rely on a single activation function, like ReLU or tanh, between layers, such a method can restrict the model's potential in discerning diverse features. Concurrent stacked activation functions operate by integrating multiple activation functions concurrently within a singular layer and aggregating their outputs. This methodology aims to blend distinct nonlinearities, facilitating a deeper comprehension of data complexities.

For a suite of activation functions represented as $\mathrm{Act}(\cdot)$, the definition of the stacked activation function $\mathrm{Act}_{\mathrm{stacked}}(x)$ aligns with the following equation:

$$\mathrm{Act}_{\mathrm{stacked}}(x) = \sum_{i=1}^{n} a_i A(x + b_i) \tag{18}$$

where a_i and b_i signify the weight coefficients and biases for the activation function, respectively. Meanwhile, n stands for the count of activation functions integrated into the stacking process.

Given the multiscale and diverse characteristics of remote sensing images, coupled with the dynamic scale fluctuations of fire and smoke, Vanilla Head incorporates a multitiered architectural approach to accommodate scenarios with diverse intricacies.

Vanilla Head integrates multiple parallel detection branches, with RFWNet featuring four distinct scale branches and RFWNet-nano encompassing three. Designed to address targets at particular scales, these branches source features from the model's foundational structure, termed the Neck network. They subsequently experience detailed feature restructuring, regression, and classification processes.

As depicted in Fig. 11, each detection branch consists of two main elements.

1) *Feature Adaptation Layer:* This layer modifies the channel count of the provided features via a 1×1 convolution process and subsequently seizes local contextual insights through a 3×3 convolution.
2) *Regression and Classification Branches:* The regression branch estimates target bounding boxes, while the classification branch assesses whether the content inside each bounding box relates to fire or smoke.

Given the notable scale fluctuations of fire and smoke observed in remote sensing images, Vanilla Head's architecture is tailored to adaptively modify both the count and parameters of its detection branches. Through statistical analysis of fire and smoke samples across multiple scales during the training stage, the requisite subdetection modules for specific scales are identified, with corresponding parameters aptly assigned to each.

The design of Vanilla Head presents significant advantages over traditional detection network heads, especially in handling complex remote sensing fire images. First, its simplified network structure reduces model complexity and parameter count, speeding up both training and inference processes, while lowering the risk of overfitting and enhancing the model's capability to handle large-scale remote sensing data. Second, through deep training strategies and stacked activation functions, Vanilla Head effectively utilizes rich nonlinear features in the early stages of training, gradually transitioning toward simple identity mappings as training progresses, thereby improving adaptability and precision in complex fire scenarios. In addition, the multibranch structure design enables the model to effectively detect fires at different scales, enhancing detection accuracy and model flexibility. These optimizations make Vanilla Head particularly suitable for remote sensing fire applications requiring rapid and accurate detection.

IV. Experimental Results and Discussion

This section presents a quantitative and visual comparative analysis of the proposed RFWNet with contemporary mainstream detection algorithms based on several evaluation metrics on the hybrid dataset presented in this article. To ensure a balanced comparison, all algorithms undergo evaluation with uniform hardware and software environments. The structure of this section is as follows. Section IV-A delves into the experimental parameters and evaluation metrics; Section IV-B contrasts the YOLO series with RFWNet in terms of forest wildfire detection from remote sensing; Section IV-C contrasts the efficacy of various object detection methods against RFWNet in detecting forest wildfires from remote sensing data; and Section IV-D showcases visual analysis of detection outcomes from different algorithms across multiple datasets.

A. Experimental Setup

1) Experimental Parameters: Our experiments operated on the Ubuntu 22.04, complemented by a GPU platform equipped with GeForce RTX 3090. Aiming for model generalizability, we implemented strategies such as random cropping, rotation, and color transformations. The experiment's learning rate stood at 0.00261, with a total of 100 iterations. Should the validation loss stagnate across ten successive cycles, the learning rate was halved. With a batch size of 32, the SGD optimization approach was adopted.

2) Comparison Algorithms: To provide an in-depth assessment of our proposed model for remote sensing forest fire detection and to distinguish its capabilities from other algorithms, we embarked on comparative experiments encompassing single-stage, two-stage, and transformer-based object detection methodologies. Such a comprehensive evaluation underscores the robustness and reproducibility of our findings.

The evaluated algorithms broadly fall into two primary categories.

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1) *YOLO Series:* Recognized as a prominent single-stage object detection algorithm, the YOLO series is lauded for its real-time performance and commendable accuracy. For a holistic comparison, we selected versions ranging from YOLO v3 to YOLO v8 [51], [52] [59], [84].
2) *Additional Object Detection Methods:* We select several representative object detection algorithms, each featuring distinct backbone networks, such as faster R-CNN [48] and RetinaNet [85] utilizing ResNet50, SSD employing VGG16, as well as RT-DETR [86], Swin, and Deformable DETR, all based on the ResNet50 backbone network and incorporating transformer technology. These choices encompass classic convolutional methods and cutting-edge techniques that emphasize advancements in training strategies and architectural design. Through these comparisons, we aim to gain a deeper understanding to the advantages our proposed RFWNet against a variety of established and advanced object detection frameworks.

B. Evaluation Metrics

1) *#Params (Number of Parameters)* indicates the total number of parameters in the model, usually expressed in millions (M). This metric reflects the scale and complexity of the model, the more parameters, the more complex the model, typically requiring more computational resources and storage space. In resource-constrained situations, it is more appropriate to prioritize models with fewer parameters.
2) *Mean Average Precision (mAP):* It serves as a prevalent evaluation metric within the realm of object detection, signifying the mean precision across diverse categories. This metric assesses the model's predictive prowess across a spectrum of thresholds, providing an all-encompassing assessment

$$\text{mAP} = \frac{1}{C_{\text{num}}} \sum_{c_{\text{cls}}=1}^{C_{\text{num}}} AP(c_{\text{cls}}) \tag{19}$$

where C_{num} is the number of categories and AP(c_{cls}) represents the average precision of category c_{cls}.
3) $\mathbf{AP_{fire}}$ and $\mathbf{AP_{smoke}}$ denote the average precision for fire and smoke categories, respectively, highlighting the model's detection performance in detecting fire and smoke instances.
4) *Precision and Recall:* Precision elucidates the fraction of true positives within the identified positives, while Recall conveys the fraction of true positives accurately identified

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{20}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{21}$$

where TP stands for true positives, FP represents the false positives, and FN denotes the false negatives.
5) AP_s, AP_m, and AP_l serve as metrics evaluating detection efficacy for small (area under 32^2 pixels), medium (area spanning 32^2–96^2 pixels), and large objects (area exceeding 96^2 pixels), respectively.
6) *Frames Per Second (FPS):* A metric measures the processing speed of the model, indicating how many frames of images the model can process per second. The higher the FPS value, the stronger the real-time processing capability of the model, indicating it is more suitable for application scenarios that require high real-time performance, such as video monitoring analysis.

TABLE I
ABLATION EXPERIMENTAL RESULTS

DCN Backbone	DDSA	Vanilla head	mAP	AP(fire)	AP(smoke)
×	×	×	64.5	65.6	63.4
✓	×	×	66.7	66.2	67.2
✓	✓	×	68.3	67.7	68.9
✓	✓	✓	69	68.4	69.6

C. Ablation Experiments

To gain a deeper understanding to the contributions of each component of our RFWNet model, we conduct a series of ablation experiments, as shown in Table I. In this section, we will analyze the results of the ablation experiments in detail to reveal the specific contributions of the DCN Backbone, dual dynamic sparse attention mechanism (DDSA), and Vanilla Head to the model's performance.

1) Baseline Model: Initially, the input image size is set to 448 × 448, and the baseline model is YOLO v8n, which does not include DCN Backbone, DDSA, or Vanilla Head. The mAP of this baseline model is 64.5%, with AP (fire) at 65.6% and AP (smoke) at 63.4%. Although this baseline model can provide certain detection effects, its performance is limited compared with more advanced configurations.

2) DCN Backbone: Upon introducing the DCN Backbone, the model's mAP significantly increased to 66.7%, with AP (fire) and AP (smoke) also increasing to 66.2% and 67.2%, respectively. This demonstrates that the introduction of the DCN backbone significantly enhances the model's ability to capture the features of flames and smoke in remote sensing images, especially when dealing with targets of varying shapes and sizes, allowing for more accurate location and recognition.

3) DCN Backbone + DDSA: Further integrating the DDSA leads to an even greater improvement in performance, with mAP rising to 68.3%, and AP(fire) and AP(smoke) reaching 67.7% and 68.9%, respectively. The addition of DDSA allows the model not only to capture the spatial features of the targets but also to adjust the allocation of attention dynamically based on the changing characteristics of the targets, thus effectively improving the accuracy of flame and smoke detection in the complex remote sensing image scenarios.

4) DCN Backbone + DDSA + Vanilla Head: Finally, with the inclusion of the Vanilla Head, the model achieves optimal performance, with mAP increasing to 69%, and AP(fire) and AP(smoke) rising to 68.4% and 69.6%, respectively. It can be concluded that the introduction of the Vanilla Head optimizes the model's classification and regression capabilities, especially in multiscale target detection scenarios, allowing for more precise identification and location of targets, which

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TABLE II
EXPERIMENTAL RESULTS OF THE YOLO SERIES ALGORITHMS FOR IMAGES OF VARYING SIZES

Image size	Model	#Params(M)	mAP(%)	$AP_{fire}(\%)$	$AP_{smoke}(\%)$	Precision(%)	Recall(%)	FPS_{4080}	FPS_{NX}
448*448	YOLO v3 tiny	12.1	62.6	60	65.2	69	51.6	769	37
	YOLO v3	62.7	66.4	62.8	70	74.6	58.1	164	5
	YOLO v3 spp	63	66.3	62.3	70.2	72.8	59.8	161	5
	YOLO v5 n	1.9	65.9	66	65.8	71.6	55.5	434	32
	YOLO v5 s	7.2	68.4	68.4	68.3	73	58.6	384	27
	YOLO v5 m	21.2	69	69.2	68.8	71.3	60.8	278	14
	YOLO v6 n	4.3	64.5	61.6	67.4	69.4	57.2	450	33
	YOLO v6 s	15	63	62.1	63.9	68.8	56.9	372	25
	YOLO v6 m	34.3	64	62.7	65.3	69.1	58.3	257	13
	YOLO v7 tiny	6.2	64	61.5	66.4	73.7	55.5	390	27
	YOLO v7	36.9	69.8	69.3	70.4	73.2	62.9	172	5
	YOLO v8 n	3	64.5	65.6	63.4	69.8	57.7	454	30
	YOLO v8 s	11.2	65.7	61.2	70.2	72.8	58.4	400	23
	YOLO v8 m	25.9	66.8	62.9	70.7	73.5	59.2	277	13
	YOLO v8 l	43	66.8	62.8	70.7	74	59	196	8
	RFWNet-nano	2	67.3	69.5	65.1	72.6	59.1	432	29
	RFWNet	8.7	69	68.4	69.6	73.9	59.3	307	20
832*832	YOLO v3 tiny	12.1	64.5	61.9	67.1	57.2	65.9	667	16
	YOLO v5 n	1.9	69.1	67.9	70.2	72.9	59.5	400	20
	YOLO v5 s	7.2	70.6	69.3	71.9	73.2	62.1	357	11
	YOLO v5 m	21.2	70.7	70	71.3	74.5	62.4	204	5
	YOLO v6 n	4.3	67.1	67.2	67	71.3	58.1	403	20
	YOLO v6 s	15	66.4	66.7	66.1	71.6	60.3	347	12
	YOLO v6 m	34.3	67.1	67.8	66.4	72.2	60.9	186	5
	YOLO v7 tiny	6.2	66.2	67.1	65.3	71.5	59	362	8
	YOLO v7	36.9	69.9	71.1	68.7	73.6	61.8	108	3
	YOLO v8 n	3	68.4	67.3	69.5	73.8	60	416	16
	YOLO v8 s	11.2	70.1	68.9	71.3	74.5	61.5	357	9
	YOLO v8 m	25.9	70.8	72.3	69.4	71.3	63.6	185	5
	YOLO v8 l	43	70.8	72.6	69.1	71.3	64	119	3
	RFWNet-nano	2	68.2	64.5	71.9	73.9	57.7	410	16
	RFWNet	8.7	71	69.4	72.6	73.5	62.5	262	7

is particularly important for the detection of flames and smoke of various sizes and shapes in remote sensing images.

D. Comparison With YOLO Series Detection Algorithms

The YOLO series has set a standard in object detection, due to its exceptional real-time detection capabilities. In this section, we compared several algorithms from the YOLO series with the model proposed in this article, as shown in Table II. To delve deeper into model performance variations with varying image dimensions, we selected images with resolutions of 448 × 448 and 832 × 832.

From Table II, it can be observed that the YOLO series of algorithms and our proposed RFWNet and RFWNet-nano models exhibit exceptional performance in both aspects highlighted. Notably, the RFWNet-nano model has only 2 million parameters, significantly reducing the demand for computational resources, while achieving high frame rates of 432 FPS with 448 × 448 image resolution on NVIDIA RTX 4080 and 29 FPS on NVIDIA Jetson Xavier NX, demonstrating its capability to satisfy real-time processing needs while maintaining high-precision detection.

Similarly, even our RFWNet model, with a modest 8.7 million parameters, far below the larger models in the YOLO series, such as YOLO v3 and YOLO v8 l (with 62.7 million and 43 million parameters, respectively), still performs remarkably well, especially with an image resolution of 832 × 832, reaching an mAP of 71% and a frame rate of 262 FPS. This proves that we effectively balance performance with computational resource consumption by optimizing the network structure and algorithm, further confirming the real-time applicability of our models in practical scenarios.

For an input image dimension of 448 × 448, YOLO v5 m achieved the mAP of 69%. This performance can be credited to the advancements in the YOLO v5 series, particularly its efforts in multiscale feature fusion, which aptly identifies targets within medium-small scale fire images. Interestingly, RFWNet mirrored this performance with an mAP of 69%. This can be ascribed to RFWNet's foundation on the DCNv3 backbone, which excels in multiscale target detection, particularly discerning the extended associations between fire and smoke. Conversely, RFWNet-nano posted an APfire value of 69.5%, outperforming other models. This superior performance stems from its DDSA mechanism, amplifying the model's precision in identifying fire within compact images. Regarding smoke, its considerable size, dynamic shape, and resemblances to the background pose detection challenges. Within this dimension, YOLO v8 l recorded the peak AP value of 70.7%. This can be attributed to YOLOv8 being the most recent addition to the YOLO series, incorporating various advanced technologies to boost its detection capabilities, thereby excelling in identifying smoke patterns.

With increased image dimensions, our RFWNet achieved a peak mAP of 71%, deriving strength from its advantage in identifying large-scale targets, notably fire. and smoke within remote sensing imagery. YOLO v8 l trails by a mere 0.2%. However, our model excels in context interpretation and compactness, primarily attributed to the incorporation

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TABLE III
COMPARATIVE EXPERIMENTAL RESULTS WITH OTHER OBJECT DETECTION ALGORITHMS

Model	Backbone	#Params(M)	mAP(%)	AP_s(%)	AP_m(%)	AP_l(%)	FPS_{4080}
Cascade-rcnn	ResNet50	23.5M	67.8	16.8	30.1	43.9	25
RTDETR	ResNet50	42.0M	71.5	19.8	32.9	48.8	22
SOLO v2	ResNet50	25.0M	69.9	13.9	30.1	44.3	33
Sparse-rcnn	ResNet50	25.8M	65	12.2	24.5	41.7	36
SSD	VGG16	143M	70.2	14.3	30.1	45	33
Swin	SwinTransformer	113.8M	70.8	18.7	32	48.2	35
Deformable DETR	ResNet50	40.0M	71.6	20.3	33.3	49.5	32
DETR	ResNet50	60.0M	70.2	19.2	32	48.3	37
FCOS	ResNet50	27.6M	68.1	17.1	30.3	46.5	28
Faster-rcnn	ResNet50	28.8M	68.5	13.6	29.3	38.7	31
Faster-rcnn	ResNet101	60.0M	69.8	13.8	29.3	39.9	22
Retinanet	ResNet50	28.6M	66	7.3	28.4	39.9	29
Retinanet	ResNet101	46..0M	67.5	7.6	29.2	43.5	19
RFWNet-nano	CSPDarknet53	2M	68.2	14.8	30.1	43.1	410
RFWNet	CSPDarknet53	8.7M	71	18.2	31.6	47.3	262

of the innovative backbone and DDSA mechanism. Given an input image dimension of 832 × 832, the model can discern the intricacies and morphology of fire with enhanced clarity. RFWNet registers an impressive APfire score of 69.4%, outperformed solely by YOLO v8 m and YOLO v8 l. While the model elucidates the morphology and structure of smoke with greater clarity, distinguishing it from the background continues to pose a challenge. YOLO v8 l attains an AP score of 69.1%, whereas RFWNet surpasses this, achieving an AP of 72.6%. This enhanced performance owes to the synergy of its foundational network and the attention mechanism, enhancing its capability to differentiate smoke from analogous backgrounds. In terms of precision and recall, particularly for images sized 832 × 832, RFWNet-nano showcased exemplary precision and recall figures relative to other models. This suggests that our model excels in accurately discerning genuine fire scenarios in real-world applications, concurrently minimizing false alarms. Yet, it merits attention that the recall rate for YOLO v3-tiny stood at 65.9% for images of 832 × 832 dimensions, in contrast to RFWNet's 62.5%. This disparity stems from the introduction of the DDSA mechanism, designed to discern the dynamic contours of fire and smoke. Consequently, the model might excessively concentrate on particular features at times, resulting in overlooked genuine target attributes. In a holistic view, given the performance enhancement of the RFWNet series, such a trade-off is deemed acceptable. Based on the evaluations across two distinct input image dimensions, both RFWNet and RFWNet-nano evidently showcased strong performance, on pair with the recent iterations of the YOLO series, and in some respects, exceeded their capabilities. Particularly, with medium–small image dimensions, our model exhibited an acute emphasis on fire and smoke, paired with proficient detection proficiencies. For larger image dimensions, they further underscored their strengths in discerning expansive targets, intricate details, and the overall context.

E. Comparative Experiments With Other Object Detection Algorithms

In this article, we conduct an in-depth comparison between RFWNet and its lightweight version RFWNet-nano against SOTA target detection models, as shown in Table III. The algorithms chosen for comparison include CNN methods based on different backbone networks and innovative methods utilizing Transformer architectures, aiming to showcase the superior performance and efficiency of the RFWNet series in remote sensing forest fire detection tasks.

It is worth noting that our model's backbone network, utilizing the CSPDarknet53 architecture, provides a robust and efficient foundation for feature extraction for the RFWNet series. Compared with other models employing ResNet50, VGG16, and SwinTransformer architectures, CSPDarknet53 maintains a low parameter count (merely 2M and 8.7M) while achieving high mAP performance (68.2% and 71%). In contrast, Deformable DETR and RTDETR, although achieving similar mAPs (71.6% and 71.5%), have significantly higher parameter counts of 40.0M and 42.0M, respectively. This demonstrates that even with fewer parameters, our optimized network structure and algorithm design can match or exceed the performance of models using heavyweight backbone networks. Furthermore, RFWNet outperforms other algorithms like Cascade R-CNN, SOLO v2, and those based on ResNet50 in detecting small (AP_s), medium (AP_m), and large (AP_l) scale targets.

In terms of accuracy, RFWNet and RFWNet-nano show outstanding performance in the detection accuracy of targets of all sizes (AP_s, AP_m, and AP_l), particularly in processing small-sized targets (AP_s), where RFWNet-nano surpasses most algorithms based on convolutional networks, thanks to our proposed DDSA, which significantly enhances the model's ability to recognize small-sized targets.

Regarding real-time capabilities, the FPS performance of our RFWNet series stands out on NVIDIA RTX 4080, with RFWNet-nano achieving 410 FPS and RFWNet reaching 262 FPS, far surpassing most comparison models. This high frame rate verifies that our RFWNet series is particularly suited for real-time remote sensing image monitoring and analysis tasks. Moreover, the low parameter design makes RFWNet-nano especially suitable for deployment on resource-constrained edge devices, offering new possibilities for fast, accurate real-time remote sensing fire detection.

Looking at the network structures, CNN-based methods (such as cascade R-CNN, SOLO v2, and faster R-CNN) rely on the deep convolutional layer stacking and complex

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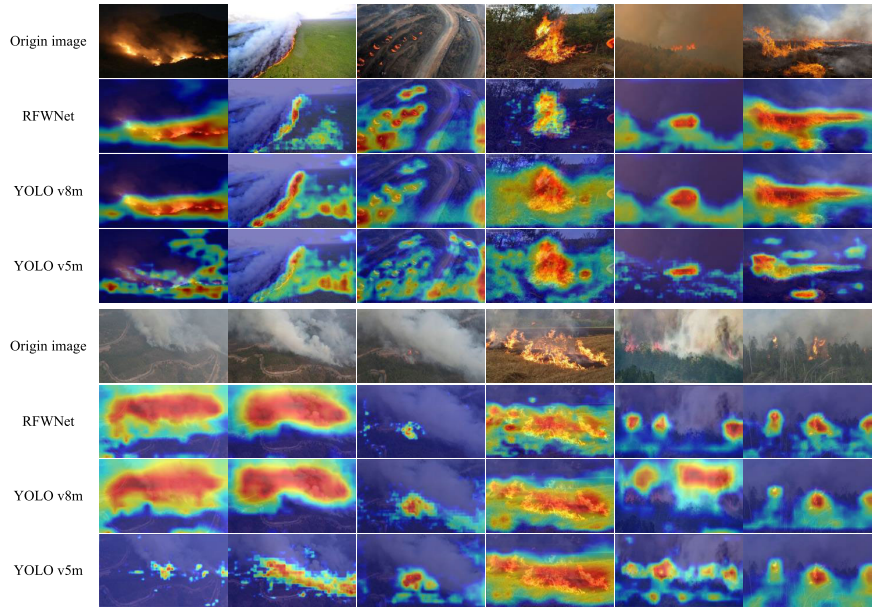


Fig. 12. Grad-CAM generated heatmaps for fire targets utilizing YOLO v5m, YOLO v8m, and RFWNet.

feature fusion strategies to enhance detection performance, which often leads to an increase in parameter count and a decrease in processing speed. In addition, Transformer-based methods (like Swin and deformable DETR) capture long-range dependencies in images through global attention mechanisms, optimizing the detection accuracy of large-sized targets. However, relative to the RFWNet series, their real-time processing capabilities are somewhat lacking, particularly in terms of parameter count and FPS.

In summary, through comprehensive comparative analysis of different algorithms, our RFWNet and RFWNet-nano not only demonstrate outstanding accuracy in the task of remote sensing forest fire detection but also showcase a clear advantage in maintaining real-time processing capabilities. These results fully validate the advanced design of the RFWNet series and its efficiency in practical applications.

F. Visualization of Experimental Results

For a clearer, more intuitive understanding of the detection algorithm's performance, this section presents visual representations of our experimental results. This includes both the precision of detection bounding boxes and the efficacy of multiple algorithms on varied datasets. Based on the experimental outcomes of Sections IV-C and IV-D, we compare the attention maps of YOLO v5m, YOLO v8m, and RFWNet, and conduct visualization experiments contrasting YOLO v5m, YOLO v8s, and YOLO v8m.

1) Visualization of Attention Maps Experimental Results: In the research of remote sensing imagery for fire detection, a model's interpretability is a key factor in assessing its performance. Gradient-weighted class activation mapping (Grad-CAM), as a popular visualization technique, can reveal the regions that a model focuses on during decision making. Therefore, we utilize Grad-CAM to generate attention heatmaps for RFWNet and other advanced models, such as YOLO v8m and YOLO v5m, in the task of remote sensing fire detection, thereby evaluating and comparing their focus on fire and smoke targets and their ability to suppress background noise.

As illustrated in Fig. 12, the heatmaps generated by Grad-CAM visually demonstrate the areas of focus for various models. The heatmaps reveal that our RFWNet focuses more precisely on the target areas compared with YOLO v8m and YOLO v5m, especially when dealing with microscale fire features such as flames. Furthermore, our RFWNet displays a high sensitivity to the areas affected by fire, which can be attributed to the effectiveness of the DDSA within the model. This mechanism allows the network to dynamically adjust attention weights, and thereby focusing on key features while ignoring irrelevant environmental information.

Notably, our RFWNet demonstrates a more balanced allocation of attention for targets of varying sizes. Even though microscale targets may occupy a minimal area in remote sensing images, RFWNet is still able to accurately capture and amplify the features of these critical regions through its refined attention mechanism. Furthermore, for larger-scale targets, RFWNet maintains its focus on the entire fire-affected area without being distracted by background noise or other interferences, indicating its excellent adaptability and robustness in recognizing targets of diverse scales.

In summary, the experimental results affirm the superior performance of our RFWNet in the task of fire detection for remote sensing imagery. In contrast, YOLO v8m and YOLO v5m, although capable of focusing on target regions in certain situations, tend to have a more dispersed distribution of attention, sometimes dedicating excessive attention to nontarget areas such as roads or vegetation. This dispersed focus can lead to diminished detection performance, particularly under complex background conditions.

2) Visualization of Synthetic Data for Experimental Results: We selected eight synthetic images for our experiments to illustrate transitions from daytime to nighttime, changes in

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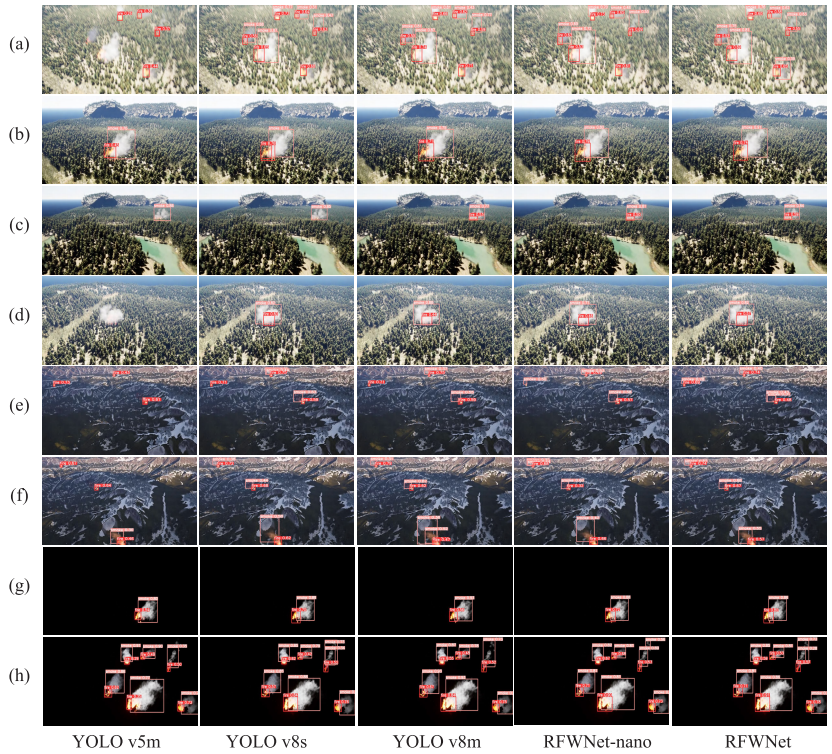


Fig. 13. Visualization of experimental results on synthetic data. (a) Simulates the dynamic blurring effect observed in real-world remote sensing images. (b) and (c) Daytime single fire target detection. (d) Daytime fire detection in synthetic data. (e) and (f) Snowy scenarios. (g) and (h) Nighttime scenes.

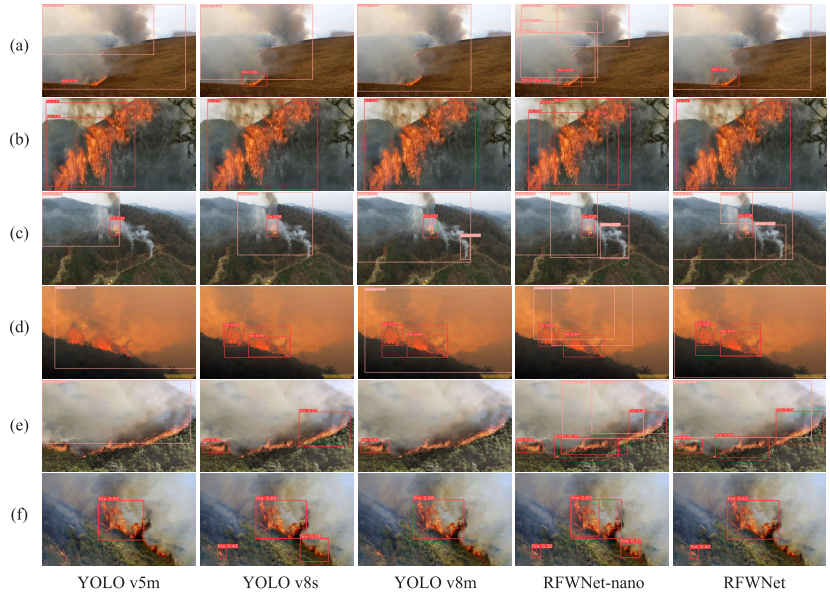


Fig. 14. Visualization results of experiments on publicly accessible fire datasets. (a) Forest fire captured from a drone perspective against a grassland background. (b) Large-scale forest fire image with a forest background. (c), (e), and (f) Multi-target forest fire images from a low-altitude drone perspective. (d) Larger-scale multi-target forest fire image with background interference.

scene complexity ranging from individual to multiple fire targets, and varying meteorological conditions spanning from clear to snowy skies, as depicted in Fig. 13. In remote sensing, fires typically manifest as small objects, presenting substantial challenges for detection algorithms. This is particularly challenging against complex backgrounds like dense forests and varied terrains. Under these conditions, RFWNet and RFWNet-nano's designs clearly improve their ability to perceive fire and smoke. This is particularly noticeable in Fig. 13(a), (c), and (d), where the RFWNet series precisely annotates fire and smoke, while the YOLO series demonstrates inconsistencies, missing some detections. In Fig. 13(a), RFWNet was the sole algorithm to precisely detect every instance of fire and smoke. In Fig. 13(c), YOLO v5m and YOLO v8s both missed certain fire instances. For the snowy scenarios in Fig. 13(e) and (f), algorithms from the YOLO series showed inconsistencies, with both missed and redundant detections. With fewer parameters and just three detection

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Fig. 15. Visualization results of experiments on Yuxi Forest wildfire data. (a)–(d) Feature images containing only smoke targets. (e)–(g) Display large-scale smoke in a forest background with flames obscured by smoke. (h)–(l) Depict forest fire images under various heights and backgrounds.

heads, RFWNet-nano exhibited limited sensitivity toward small-scale targets, overlooking some small fire instances. In contrast, RFWNet precisely identified all fire and smoke targets. For the darker scenes in Fig. 13(g) and (h), we employed data augmentation techniques on fire and smoke, leading to pronounced contrasts against the background. In summary, RFWNet and RFWNet-nano consistently display exemplary fire detection prowess in a variety of challenging scenarios. Relative to other algorithms, their performance is distinguished by notable accuracy and consistency.

3) Visualization of Experimental Results on Publicly Available Fire Data: The public fire dataset consists of images taken from a ground angle, supplemented by a few captured from a remote sensing perspective, with most of these images being of a relatively low resolution. We chose six images for assessing the model's performance in identifying fire targets in real-world scenarios, as illustrated in Fig. 14. Fig. 14(a) displays a fire juxtaposed against a grassland background, covering fire targets of multiple sizes and more prominent smoke targets. YOLO v5m missed the evident fire targets and identified smoke targets excessively. YOLO v8m failed to identify the fire targets. RFWNet-nano precisely identified fire targets of different sizes, yet it overly detected the smoke targets. Given the reduced resolution of Fig. 14(b) and (f) and the smoke's resemblance to the background, all algorithms struggled to precisely identify the smoke targets. In Fig. 14(c), neither YOLO v5m nor YOLO v8s fully identified the smoke targets. In Fig. 14(d), YOLO v5m failed to detect the fire, whereas YOLO v8s did not recognize the smoke. In Fig. 14(e), YOLO v5m, YOLO v8s, and YOLO v8m all exhibited varying degrees of detection omissions. In summary, both RFWNet and RFWNet-nano succeeded in precisely identifying all fire targets, surpassing the other algorithms in the evaluation. Though detection redundancy was noted in specific images, this arises from the fire and smoke's varying forms, leading the model to perceive segments of the fire or smoke as fire targets.

4) Visualization of Experimental Results on Yuxi Forest Wildfire Data: We selected 12 images depicting the Yuxi

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wildfire, which capture fire and smoke from diverse altitudes. Importantly, because of the fire's significant scale, the smoke was pronounced, with fires appearing small and frequently hidden by smoke. Fig. 15 displays the visualized experimental results. Fig. 15(a)–(d) primarily depict smoke. Considering the extensive presence of smoke, most algorithms successfully detected it. However, YOLO v8s overlooked certain smoke targets and incorrectly labeled trees and ground as fire in Fig. 15(d). Fig. 15(e)–(g) exhibit vast smoke against a forested background, with fires hidden by the smoke. Under such conditions, YOLO series algorithms displayed detection shortcomings, while both RFWNet and RFWNet-nano precisely detected a majority of the smoke. Remarkably, RFWNet identified fires that were nearly invisible to the unaided eye. Fig. 15(h)–(l) represent forest fires viewed from different altitudes. Compared with the YOLO series algorithms, solely RFWNet and RFWNet-nano recognized every fire-related target. In conclusion, both RFWNet and RFWNet-nano consistently showcased remarkable detection capabilities across various Yuxi wildfire situations. Especially in identifying obscured or tiny fires, they markedly exceled over other methods, highlighting their robustness and accuracy in complex real-world fire scenarios.

V. Conclusion

This article chiefly focuses on the real-time detection of forest fires via remote sensing, presenting RFWNet along with its lightweight version, RFWNet-nano. Addressing the limited availability of remote sensing data, we adopted a DT approach to create a synthetic dataset. We complemented this with on-site forest wildfire data from Yuxi city, gathered through drones, and made publicly accessible on Github. In response to the challenges stemming from the dynamic nature of fires and smoke and their extensive correlations, our RFWNet integrates a multigroup backbone network and a DDSA mechanism, boosting the model's perceptual strengths. The Vanilla Head endows the model with adaptability in recognizing targets across different scales. In comparative evaluations against multiple algorithms, RFWNet showcases heightened accuracy and resilience. In future endeavors, we aim to explore the model's adaptability across multiple scenarios and investigate advanced hardware acceleration techniques to cater to intricate situations.

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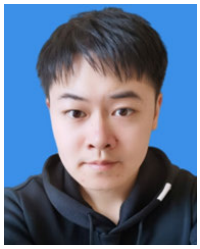


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